

March 31, 2026

Editor-in-Chief
IEEE Transactions on Information Theory

Dear Editor-in-Chief,

I submit the manuscript “Exact Recovery Under Deterministic Partial Views: Confusability Graphs, Strong Powers, and Capacity” for consideration as a regular paper in the IEEE Transactions on Information Theory.

Unlike the classical zero-error setting where the observation law is given, here a deterministic partial-view architecture produces the confusability graph. A family of admissible coordinate-subset views induces a confusability graph on latent states; exact recovery becomes equivalent to graph colorability; repeated composition lifts the one-shot graph to its strong powers; and the resulting normalized block rates converge to the Shannon capacity of the induced graph. On the full labeled tuple space, the realizable confusability relations are characterized exactly: they are precisely those determined by upward-closed families of coordinate-agreement sets. The capacity bounds inherit the standard Lovász- ϑ upper theory, applied to the model-induced graph class.

The paper also identifies a structural equality route: when confusability is transitive, the induced graph collapses to a cluster graph, yielding capacity- ϑ equality, with meet-witnessing and fiber coherence as sufficient conditions.

The later sections show the theory is non-vacuous: nearly all widely deployed deterministic partial-view architectures (spanning language runtimes, database engines, and dependency managers) fall into the non-trivial regime above the zero-incoherence boundary, where non-clique failure topologies and bounded exact-success rates apply.

All principal theorems are machine-checked in Lean 4. A supplementary artifact containing the formalization is available at <https://doi.org/10.5281/zenodo.18141365>.

Thank you for your time and consideration.

Sincerely,

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